
PYTHON STRING METHODS

◆ 1 Case Conversion Methods

👉 Used to change the case (capital/small letters) of a string.

Method	Syntax	Parameters	Detailed Explanation
upper()	s.upper()	None	This method converts all letters of the string into uppercase letters and returns a new string.
lower()	s.lower()	None	This method converts all letters of the string into lowercase letters and returns a new string.
capitalize()	s.capitalize()	None	This method converts the first character of the string into uppercase and all remaining characters into lowercase.
title()	s.title()	None	This method converts the first character of each word in the string into uppercase and is mainly used for formatting headings.
swapcase()	s.swapcase()	None	This method converts uppercase letters into lowercase and lowercase letters into uppercase.

◆ 2 Searching & Counting Methods

👉 Used to find or count characters and substrings in a string.

Method	Syntax	Parameters	Detailed Explanation
find()	s.find(sub[, start[, end]])	sub – substring	This method searches for a substring and returns its index position. If the substring is not found, it returns -1.

rfind()	s.rfind(sub[, start[, end]])	sub – substring	This method searches for a substring from the right side of the string.
index()	s.index(sub[, start[, end]])	sub – substring	This method returns the index of the substring but raises an error if the substring is not found.
count()	s.count(sub[, start[, end]])	sub – substring	This method returns the total number of occurrences of a substring in the string.

◆ 3 Checking / Validation Methods

👉 Used to check the type of characters present in the string.

Method	Syntax	Parameters	Detailed Explanation
isalpha()	s.isalpha()	None	Returns True if the string contains only alphabets and no digits or spaces.
isdigit()	s.isdigit()	None	Returns True if the string contains only digits.
isalnum()	s.isalnum()	None	The isalnum() method returns True if the string contains only alphabets, only digits, or a combination of alphabets and digits , and does not contain spaces or special characters
islower()	s.islower()	None	Returns True if all alphabetic characters in the string are lowercase.
isupper()	s.isupper()	None	Returns True if all alphabetic characters in the string are uppercase.
isspace()	s.isspace()	None	Returns True if the string contains only whitespace characters.

startswith()	s.startswith(prefix)	prefix	Checks whether the string starts with the given prefix.
endswith()	s.endswith(suffix)	suffix	Checks whether the string ends with the given suffix.

◆ 4 Trimming / Whitespace Removal

👉 Used to remove unwanted spaces from a string.

Method	Syntax	Parameters	Detailed Explanation
strip()	s.strip([chars])	chars (optional)	Removes spaces or specified characters from both ends of the string.
lstrip()	s.lstrip([chars])	chars (optional)	Removes spaces or specified characters from the left side of the string.
rstrip()	s.rstrip([chars])	chars (optional)	Removes spaces or specified characters from the right side of the string.

◆ 5 Splitting & Joining

👉 Used to break a string into parts or combine multiple strings.

Method	Syntax	Parameters	Detailed Explanation
split()	s.split(sep=None, maxsplit=-1)	sep – separator	Splits the string into a list of words using a separator.
rsplit()	s.rsplit(sep=None, maxsplit=-1)	sep – separator	Splits the string from the right side.
splitlines()	s.splitlines(keepends=False)	keepends	Splits the string at line breaks and returns a list of lines.
join()	'sep'.join(iterable)	iterable	Joins elements of a list or tuple into a single string.

◆ 6 Replace & Modify Methods

👉 Used to change or format string content.

Method	Syntax	Parameters	Detailed Explanation
replace()	s.replace(old, new[, count])	old, new	Replaces one substring with another substring in the string.
zfill()	s.zfill(width)	width	Adds leading zeros until the string reaches the given width.
center()	s.center(width[, fillchar])	width	Aligns the string in the center by adding padding characters.
ljust()	s.ljust(width[, fillchar])	width	Aligns the string to the left by adding padding characters.
rjust()	s.rjust(width[, fillchar])	width	Aligns the string to the right by adding padding characters.

◆ 7 String Formatting Methods

👉 Used to insert values into strings.

Method	Syntax	Parameters	Detailed Explanation
f-string	<code>f'{var}'</code>	Variable	Inserts variable values directly into a string.
format()	<code>"{}".format(value)</code>	Value	Inserts values into placeholders.
% operator	<code>"%d" % value</code>	Value	Formats strings using old-style formatting.

◆ 8 String Comparison Operators

👉 Used to compare strings alphabetically.

Operator	Syntax	Detailed Explanation
<code>==</code>	<code>s1 == s2</code>	Checks whether two strings are exactly equal.
<code>!=</code>	<code>s1 != s2</code>	Checks whether two strings are different.
<code><</code>	<code>s1 < s2</code>	Compares strings based on alphabetical order.
<code>></code>	<code>s1 > s2</code>	Compares strings based on alphabetical order.